

Cloud-Native AI Fitness Coach with Health Risk Prediction and Personalized Guidance

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2. Abstract:

In today's fast-paced digital world, maintaining physical fitness and well-being is increasingly challenging, with many individuals relying on generic and ineffective fitness solutions. This project presents a **Cloud-Native Fitness Coach** that uses cloud technologies to deliver personalized fitness guidance. Users provide inputs such as age, weight, height, activity level, and fitness goals, which are processed to generate customized workout plans and useful health insights.

Key features include:

- Personalized workout plan generation
- Basic health assessment based on user inputs
- Real-time progress tracking
- Data-driven lifestyle recommendations

The system is built using a scalable microservices architecture on Google Cloud Platform. Cloud Run is used for deployment and scalability, BigQuery handles data storage and analysis, and Cloud Storage manages user data and reports. This solution promotes accessible and efficient fitness management through cloud-based technology.

3. Sustainable Development Goals (SDGs)

SDG 3: Good Health and Well-being

SDG 10: Reduced Inequalities

4. Google Cloud Components Used

- Cloud Run
- BigQuery
- Vertex AI
- Cloud Storage